

Assignment no. 4

Exercise 1 Static vs Contextual Embeddings

In this assignment, you will explore the differences between static word embeddings and contextual embeddings using the BERT model. You will extract embeddings from different layers, apply t-SNE visualization, and analyze how meaning shifts with context.

Tasks:

1. Load BERT tokenizer and model
2. Prepare a list of example sentences
3. Extract static embeddings (embedding layer)
4. Extract contextual embeddings (final hidden layer)
5. Run t-SNE on both sets of embeddings
6. Visualize and compare the representations prepare this for the presentation in the exercise.
7. Research if/when the static and contextual embeddings are still comparable to each other.

Deliverables: A notebook with embedding extraction, visualizations, analysis and interpretation.

Exercise 2 Semantic Search and Retrieval Using Embeddings

In this final exercise, you will build a semantic search system using multiple embedding types and evaluate retrieval quality. This demonstrates how embeddings enable downstream tasks beyond classification.

Task Description: You will implement semantic search using:

- Word2Vec (static embeddings)
- BERT mean-pooled contextual embeddings
- Sentence-BERT (SBERT) sentence embeddings

Steps:

1. Load or create a small text corpus (FAQ-like statements)
2. Generate embeddings using all three methods
3. Implement cosine-similarity-based retrieval
4. Rank documents for a user query
5. Evaluate with:
 - Precision@k
 - Recall@k
 - Mean Reciprocal Rank (MRR)
6. Visualize embeddings using t-SNE

Deliverables:

- Notebook containing all embedding methods
- Retrieval system implementation
- Evaluation metrics and t-SNE visualization
- Discussion of results for the presentation: What are the limits of this approach?

Exercise 3 Text Classification with various Embeddings

In this exercise, you will classify [SMS messages](#) as spam or ham using multiple embedding methods.

Embedding Methods:

- BERT [CLS] token embedding
- BERT mean-pooled token embeddings
- Word2Vec trained on the dataset

You will train a classifier of your choice on each embedding type and compare performance using classification reports, confusion matrices, and qualitative error analysis. Additionally research online when to use the [CLS] token and when to use mean-pooling.

Deliverables:

- Embedding extraction code
- Classifier training and evaluation
- A comparison of all methods to present in the exercise.
- Analysis on [CLS] token vs mean-pooling.

Exercise 4 Code Classification with BERT Embeddings

This exercise you will classify short code snippets using contextual embeddings. As calculating the embeddings is quite resource intensive you are provided the [embeddings](#) and the base [notebook](#) on eLearning. You will:

- Load and preprocess a labeled code dataset
- Generate BERT embeddings (already provided to you to save time)
- Train a traditional classifier (eg. LogisticRegression from sklearn)
- Evaluate with confusion matrix and classification report
- Prepare for the presentation in the exercise, what worked well / didn't work well and why that might be the case.

Deliverables: Notebook with embedding extraction, model training, evaluation, and analysis of the findings.

When using Large Language Models (eg. ChatGPT, Copilot, ...) to produce output that is handed in, also state the prompt and model used in the submission. Submissions that are not marked and are noticeably AI generated will be graded with zero points!

Submission: electronically via Moodle; Deadline: Tue, Apr 07, 2026, 23:59.